15-Minute Day-Ahead Pricing and Information Processing

Germany / DE-LU electricity market

Empirical design — models & specifications only

TOPIC 1

Overall sensitivity

Does higher market granularity change the price response to expected scarcity?

TOPIC 2

Within-hour information use

Once 15-minute day-ahead prices exist, do they reflect quarter-hour scarcity?

TOPIC 3

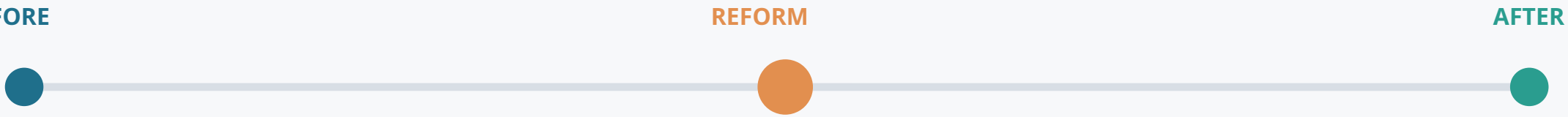
Updating to new information

Did the reform change how intraday prices adjust when forecasts are revised after day-ahead?

Market-design reform: 1 October 2025

Research design in one picture

The reform changes the timing and granularity of price formation — not the underlying delivery asset.



DA market: 1 hourly price
15-min fundamentals exist
15-min intraday trading exists

1 Oct 2025
Day-ahead MTU
60 → 15 minutes

DA market: 4 prices/hour
Same quarter-hour delivery blocks
Intraday remains granular

Topic 1 — Sensitivity

Compare the hourly-equivalent price response to forecast residual load before and after the reform.

Topic 2 — Incorporation

Within post-reform hours, test whether quarter-hours that are forecast tighter also receive higher DA prices.

Topic 3 — Updating

Use forecast revisions and intraday prices to test whether response to genuinely new information changes after the reform.

Attention → incorporation of known information → updating to new information

Behavioral framework

The paper treats the reform as a change in the information-processing environment.

Neoclassical benchmark

- Relevant information is incorporated into prices.
- New information is filtered according to its signal content.
- More price intervals should not, by itself, create systematic mispricing.

Behavioral channels

- Limited attention / cognitive load: more decisions can dilute processing.
- Salience: 15-minute signals may receive more weight once directly priced.
- Belief distortions: traders may underweight or overweight noisy updates.

Mapping to the three empirical topics

TOPIC 1

How much weight?

Aggregate sensitivity to expected scarcity

→

TOPIC 2

Is known detail used?

Within-hour incorporation of relevant information

→

TOPIC 3

How are beliefs updated?

Price response to new forecast information

Data architecture and notation

A common 15-minute panel links forecasts, day-ahead prices, intraday prices, and realizations.

SMARD — fundamentals

Forecast residual load
Day-ahead prices
DA renewable forecast
Intraday renewable forecast
Actual generation / consumption

Intraday price benchmark

Quarter-hour intraday price for the same delivery block.

Preferred benchmark: a late intraday index / trade-based price that occurs after the selected forecast update.

Panel index

q = quarter-hour delivery block
h = containing hour
d = delivery date
Post = 1 from 1 Oct 2025 onward

Core transformations

Hourly-equivalent (Topic 1)

Average the four quarter-hour observations within each hour so pre/post are compared at the same temporal scale.

Within-hour deviation (Topic 2)

Subtract the mean of the four quarters in the same hour. This removes the common hourly level.

Forecast revision (Topic 3)

Compare the DA renewable forecast with a later intraday forecast for the same delivery quarter.

Planned sample: 1 Oct 2024 – 30 Sep 2026

Topic 1 — Did aggregate scarcity sensitivity change?

A before/after slope comparison at an hourly-equivalent scale.

MODEL

$$P_{d,h}^{DA,H} = \alpha_d + \lambda_h + \beta_1 RL_{d,h}^{DA,H} + \delta_1 \left(RL_{d,h}^{DA,H} \times Post_d \right) + \varepsilon_{d,h}$$

Outcome

$P^{DA,H}$ d,h: hourly-equivalent day-ahead price.

Post-reform: average the four 15-minute DA prices inside hour h.

Main regressor

$RL^{DA,H}$ d,h: hourly-equivalent forecast residual load = forecast demand − forecast wind − forecast solar.

Key estimand

δ_1: change in price sensitivity to forecast scarcity after 1 Oct 2025.

Pre = β_1 • Post = $\beta_1 + \delta_1$

Specification

- Date fixed effects (αd): absorb day-wide price conditions.
- Hour-of-day fixed effects (λh): absorb the systematic daily price shape.
- Standard errors clustered by delivery date.
- Robustness: allow the hour-of-day profile to differ pre/post.

Behavioral interpretation

Does a more granular market change overall responsiveness to a relevant fundamental signal? This maps to competing salience versus cognitive-load channels.

Topic 2 — Do 15-minute DA prices use known within-hour information?

Post-reform mechanism test: compare relative scarcity with relative price inside the same hour.

CONSTRUCTION

$$p^{dev}_{q,h} = P^{DA}_{q,h} - \bar{P}^{DA}_{h} \qquad k_{q,h} = RL^{DA}_{q,h} - \bar{RL}^{DA}_{h}$$

MODEL

$$p^{dev}_{q,h} = \beta_2 k_{q,h} + \varepsilon_{q,h}$$

Why this specification is clean

- The transformation removes everything common to the exact hour.
- Equivalent to an individual-hour fixed-effect regression.
- Focuses only on the four quarter-hours relative to one another.
- Estimated on the post-reform period because pre-reform DA prices are mechanically flat within each hour.

Interpretation of β_2

How much higher the DA price is for a quarter-hour that is forecast to be 1 GW tighter than the average quarter in its own hour.

Behavioral interpretation

Once the market allows four separate prices per hour, do prices actually use the finer relevant information? This is an information-incorporation test, not a causal pre/post efficiency test.

Topic 3 — Did the reform change updating to new information?

Separate information known at DA from information that arrives after DA.

Known scarcity Kq

Already available at the DA stage:

Kq = RLDA,q − average RLDA within hour

News Nq

Forecast revision after DA:

Nq = Frenew,DA,q − Frenew,ID,q
Positive = renewables revised down = tighter

Intraday correction Cq

Change in within-hour price shape:

Cq = (PID,q − avg PID,h) − (PDA,q − avg PDA,h)

MAIN MODEL

$$C_{q,h} = \theta K_{q,h} + \beta N^{W}_{q,h} + \delta_3 \left(K_{q,h} \times Post\right) + \gamma \left(N^{W}_{q,h} \times Post\right) + \varepsilon_{q,h}$$

where $N^{w}q,h$ = Nq,h − average N within the same hour

θ

Before reform: how much intraday corrects quarter-hour scarcity that was already known at DA.

δ_3

Change after reform in the correction of already-known scarcity. A shift toward zero means earlier incorporation in DA.

β

Before reform: price response to genuinely new renewable forecast information.

γ — key behavioral estimand

Change after reform in the price response to genuinely new information.

Topic 3 — Timing and identification

The forecast revision must precede the intraday price benchmark used as the outcome.

D−1

Day-ahead information set

DA residual-load forecast
DA renewable forecast
DA price clears

Later forecast update

Intraday renewable forecast arrives.
News = DA forecast − later forecast.

Intraday price benchmark

Use a quarter-hour price observed after the selected forecast update for the same delivery block.

Key timing rule

15-minute forecast ≠ update every 15 minutes

The 15-minute timestamp identifies the delivery interval being forecast. It does not imply that the forecast itself is refreshed every 15 minutes.

Price benchmark choice

Candidate benchmark: Netztransparenz IP Index (ID-AEP) or another consistent 15-minute intraday index. The chosen price must come after the forecast snapshot.

Specification choices

- Within-hour demeaning absorbs exact-hour common shocks.
- Cluster standard errors by delivery date.
- Validate that the forecast revision is informative using eventual actual generation.
- Robustness: quarter-position effects and alternative intraday price benchmarks.

Unified empirical roadmap

Each topic isolates a different stage of information processing.

Topic	Question	Main estimand	Behavioral channel	Design
1	Did overall scarcity sensitivity change?	**δ_1: change in RL → price slope**	Attention / salience vs cognitive load	Pre/post hourly-equivalent
2	Do 15-minute DA prices use known quarter-hour scarcity?	**β_2: within-hour scarcity → price**	Use of relevant fine-grained information	Post-only within-hour FE
3	Did updating to new information change?	**γ: change in news → intraday correction**	Belief updating / attention to new signals	Pre/post with intraday benchmark

Core contribution

The reform lets us distinguish whether higher temporal granularity changes the amount of attention to fundamentals, the timing of information incorporation, and the response to genuinely new information.